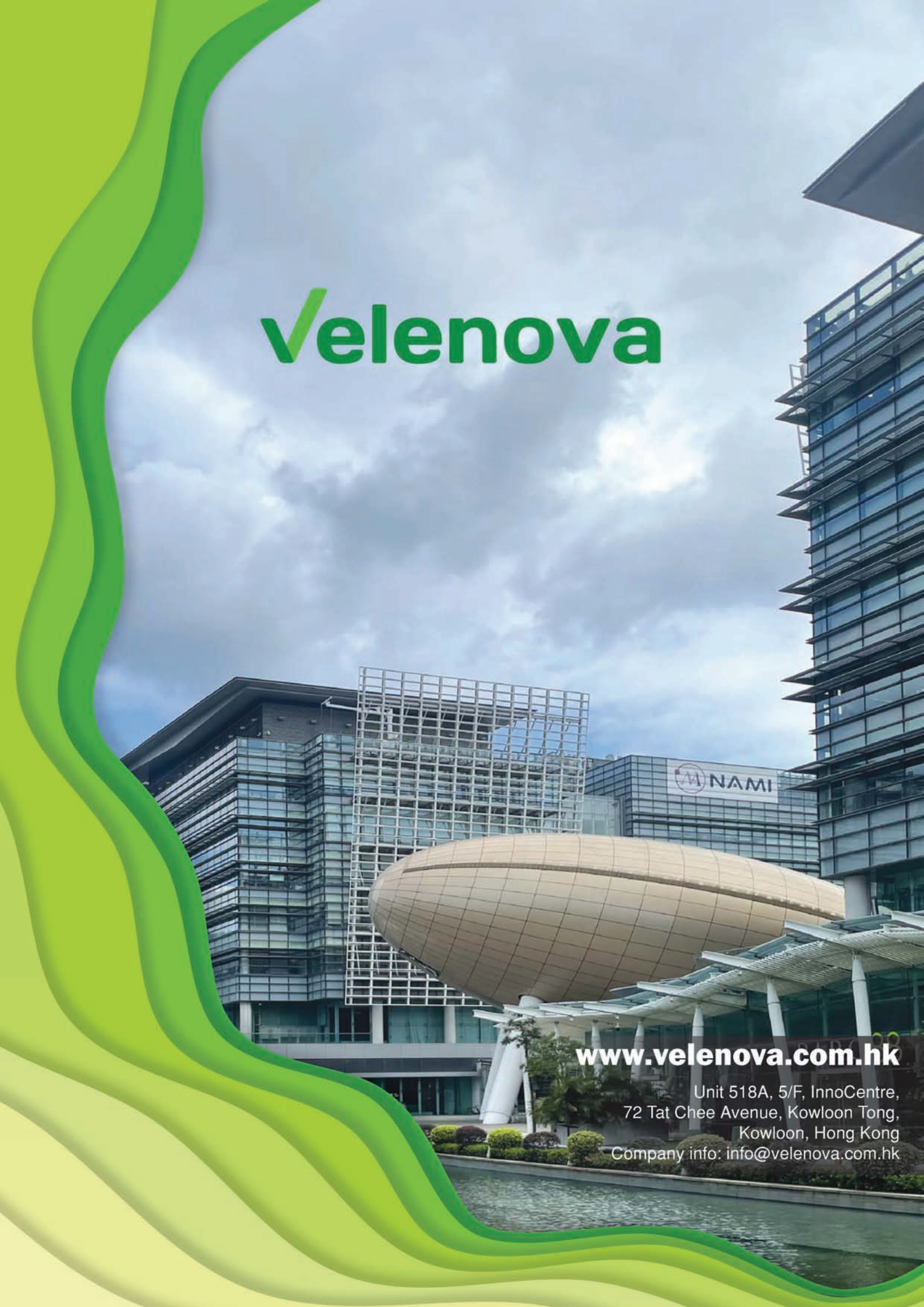


The image features a large, stylized green wavy graphic on the left side, resembling a stylized 'V' or a series of waves. The background is a photograph of a modern building complex under a cloudy sky. The building has a prominent glass facade and a large, curved, metallic structure in the foreground. A sign on the building reads 'NAMI'.

✓ velenova

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mmwpcb.com :

Your One-Stop Industrial Electronic Solutions Partner

- Comprehensive Solutions in PCB, 3D Printing & RF Testing

About Us

MMWPCB works on building a one-stop industrial electronic platform, realizing PCB manufacturing, 3D printing and RF testing. With advanced technology, strict quality control and efficient delivery, we are committed to meeting the diverse needs of industries such as electronics, telecommunications and automotive. Our integrated services cover from prototype development to mass production, supported by a team of experienced engineers, ensuring the rapid production of high-reliability and cost-effective products for every customer.

Our Manufacturing Services

PCB & Related Manufacturing

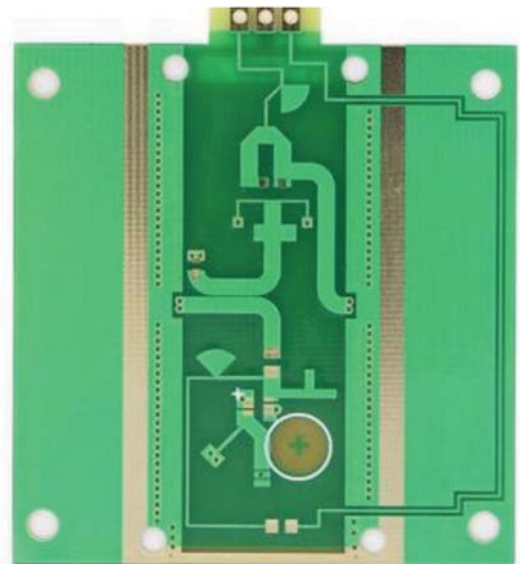
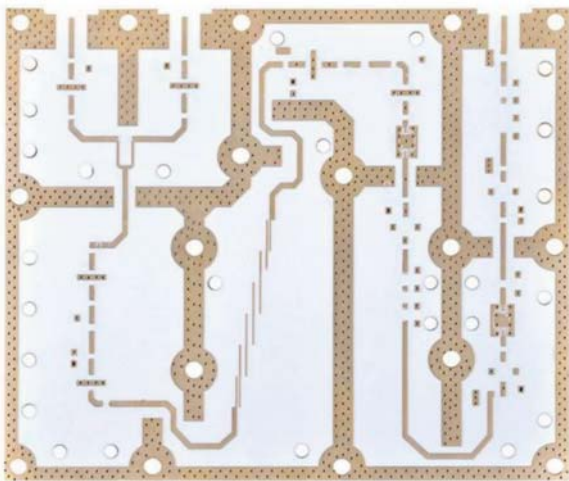
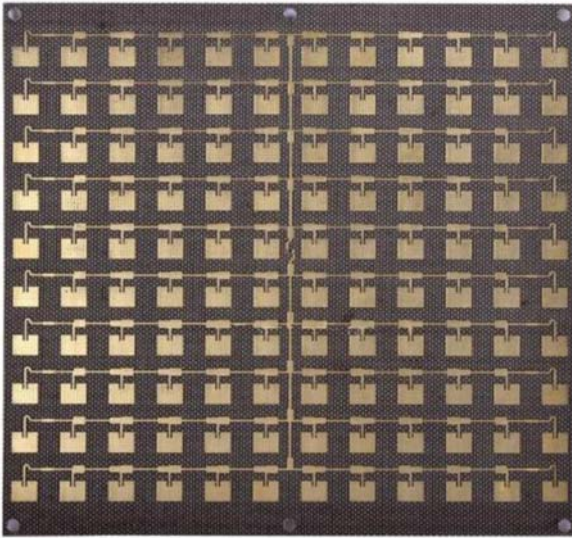
Standard PCB: We offer high-quality standard PCBs with various specifications, including FR-4 and other specific substrates. Available in different thicknesses, copper weights and layer counts. Fast turnaround options are available for prototype and small-batch orders.

Advanced PCB: Focusing on high-end applications, our advanced PCBs include high-frequency PCBs (for 5G, radar), high-temperature PCBs (resistant to extreme temperatures), and HDI (High-Density Interconnect) PCBs. Fine line widths/spaces and blind/buried vias support complex circuit designs and high-speed signal transmission

Main Process Capabilities of PCB

Substrate Type	FR4, PTFE, Hydrocarbon, Composite Dielectric, Ceramic, PI
Hole Processing Capacity	Machine Drilling (d) : Min 0.15mm, Max 6.5mm, Aspect Ratio Max 16:1
	Blind Hole (d): Min 80um Max 150um Aspect Ratio=1:1
	Min Pitch 0.25mm
	Hole Tolerance: Plating Through Hole $\pm 0.08\text{mm}$, Non Plating Through Hole $\pm 0.05\text{mm}$
Hole Copper Thickness	Hole Position Accuracy : $\pm 0.025\text{mm}$
	Through Hole : 18-25um
	Blind Hole : 18-25um
	Buried Hole : 15-25um
Line Width (W) / Line Distance (d)	HDI Hole : Laser Hole 12-15um, Blind Hole 15-25um
	Inner: Base Copper Hoz 0.075mm/0.075mm; Base Copper 1oz 0.1mm/0.1mm; Base Copper 2oz 0.15mm/0.15mm
	Outer: H oz Min 0.075/0.075mm
Line Width Tolerance	$\pm 10\%$
Distance From Wire to Edge of Board (D)	Min 0.2mm

PCB Products



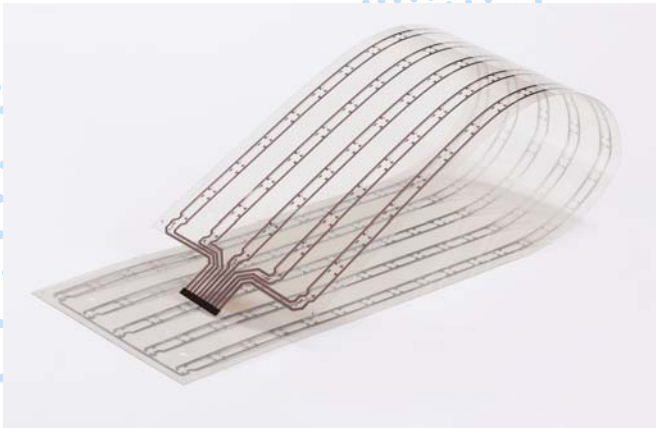
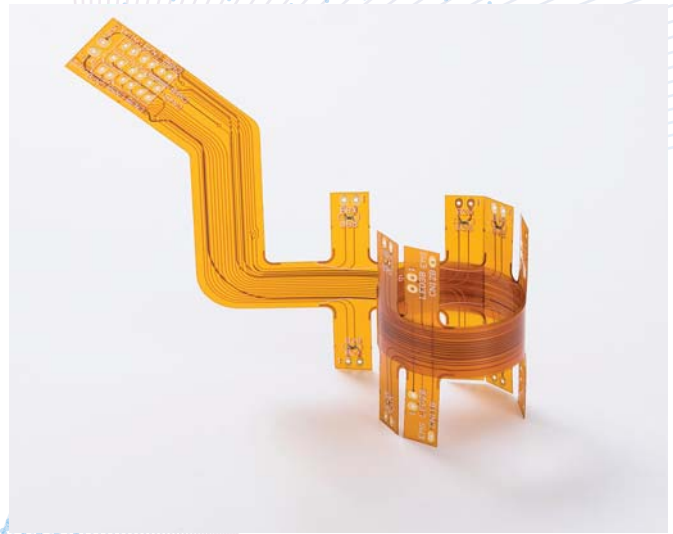
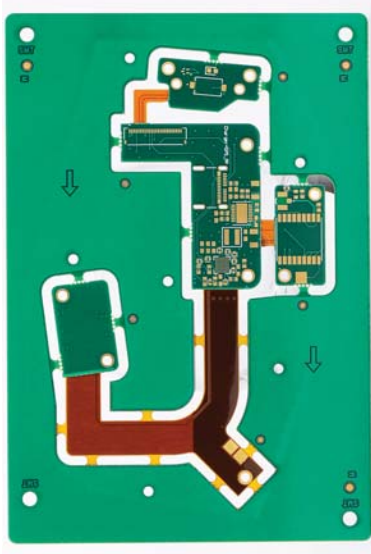
Flexible PCB: Made from flexible substrates like PI and LCP, our flexible PCBs offer excellent **bendability** and durability. They are suitable for devices with **limited space** or dynamic applications, such as wearable electronics, medical devices and automotive sensors. Customization options include materials, layer count and surface finishes.

Rigid flex PCB: Combining the advantages of rigid and flexible PCBs, rigid flex PCBs integrate **rigid boards** for component mounting and flexible **boards** for connectivity. They reduce assembly steps, improve reliability and save space, widely used in mobile communications, smart home devices, industrial control systems and medical equipment.

Main Process Capabilities of Flexible PCB and Rigid flex PCB

Layers	Flexible PCB	1-14
	Rigid flex PCB	2-16
Hole	Mechanical hole diameter	0.15-6.2mm
	Electroplating laser blind hole aperture	0.1-0.15mm (0.1mm preferred)
	Hole position accuracy tolerance (compared to CAD data)	±3mil
Line width/line spacing	Internal layer	1/30Z: 1.5/1.5 mil
		1/20Z: 2/2 mil
		10Z: 3.5/3.5 mil
		20Z: 5/5 mil
		30Z: 7.5/7.5 mil
		40Z: 10/10 mil
		50Z: 15/15 mil
	Outer layer	1/30Z: 3/3.5 mil
		1/20Z: 3.5/3.5 mil
		10Z: 4.5/5 mil
		20Z: 6/8 mil
		30Z: 8/12 mil
		40Z: 9/15 mil
	Line width tolerance	≤10mil: ±1.0mil; >10mil: ±1.5mil
Surface coating thickness	Solder Coating	2-40um (0.4um lead spray tin surface, 1.5um lead-free spray tin surface)
	Electronickel	Nickel thickness :≥3um; Gold thickness: 0.025-3um
	Nickel palladium	Nickel thickness :3-8um; Palladium thickness: 0.05-0.3um; Gold thickness: 0.05-0.1um
	Sink tin	≥0.6um
	Silvering	0.2-0.4um
	OSP	0.3-0.5um
	Green oil	10-18um(copper cover oil),5-8um (through hole cover oil)

Flexible PCB and Rigid flex PCB Products



PCB Assembly: We provide full turnkey PCB assembly services, including SMT (Surface Mount Technology) and THT (Through-Hole Technology) assembly. Equipped with advanced pick-and-place machines, reflow ovens, and automated inspection systems, we ensure precise component placement and soldering. We support BOM management, component sourcing and post-assembly testing.

SMD Stencil: Custom SMD stencils tailored to your PCB design. Suitable for both prototype and mass production assembly.

Additive Manufacturing & Molding

3D Printing: Offering a range of 3D printing technologies, we produce prototypes and functional parts with high accuracy. Materials include metals and non metals. Fast turnaround for design validation and low-volume production, supporting complex geometries that traditional manufacturing cannot achieve.

3D Printing Products



Injection Molding: For high-volume production of plastic parts, our injection molding services use advanced machines and precision molds. We handle various thermoplastic materials, ensuring consistent part quality, tight tolerances, and cost efficiency. Suitable for components in consumer electronics, automotive parts and medical devices.

Vacuum Casting: Ideal for small-batch production (10-500 units) of plastic parts with high surface finish and detail. Fast lead times compared to traditional injection molding, perfect for pre-production validation and market testing.

Precision Manufacturing

CNC Machining: Our CNC machining services cover milling, turning, and drilling of various materials, including aluminum, stainless steel, copper and carbon fiber. With high-precision equipment, we produce complex parts for enclosures, fixtures, and mechanical components, supporting both small-batch prototyping and large-scale production.

Sheet Metal Fabrication: Specializing in sheet metal cutting, bending, welding and surface treatment (painting, plating), we manufacture high-quality sheet metal parts. Materials include aluminum, stainless steel and copper, with applications in electronic enclosures, industrial equipment and structural components. We ensure consistent quality and tight dimensional control.

Professional RF Testing Services

We provide professional RF testing services, delivering accurate and industry-standard-compliant test results. Our advanced facilities support a wide range of tests, including frequency domain and time domain testing. Whether for R&D or production, our expert team delivers precise and reliable results for your products across a wide frequency range to meet diverse needs.

Core Equipment



Signal Source (100kHz-50GHz)



Spectrum Analyzer (2Hz to 50 GHz)



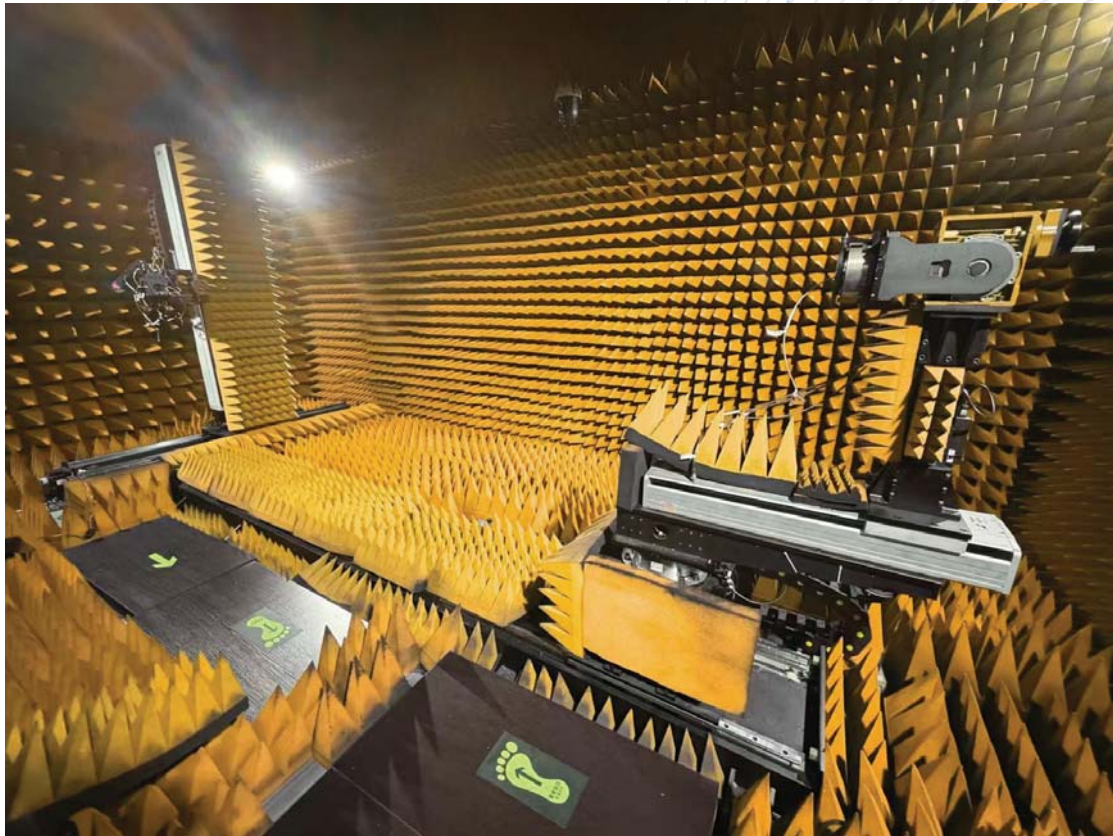
Vector Network Analyzer (10MHz to 50 GHz)



Material Characterization Kit (25GHz-40GHz)



Oscilloscope (6 GHz maximum bandwidth)



Antenna test system



mmWave/THz circuit test platform

Frequency Domain RF Test: This includes S parameter analysis (S_{11} , S_{21} , etc.) to evaluate signal transmission and reflection, spectrum-related measurements (power, harmonics, spurious emissions) for compliance with regulatory standards and material characterization (dielectric constant, loss tangent) to support accurate design

Measurement Capabilities:

(1) S-parameter Analysis

- Full S-parameter characterization up to 500 GHz including insertion loss, return loss, and group delay.

(2) Spectrum-Related Measurements:

- Frequency range from 2Hz to 50 GHz
- Frequency response includes gain and attenuation
- Harmonics, spurious and phase noise
- Passive Device Testing: filters, mixers, antennas, and mmWave ICs.

(3) Material Characterization

- Frequency range from 25 GHz to 40 GHz.
- Permittivity extraction using waveguide methods
- Permittivity error $\epsilon_r \pm 1\%$
- Loss tangent error $\tan\delta \pm 2\%$

Time Domain Measurement: We provides time domain testing using advanced oscilloscope, enabling accurate signal analysis for high-speed digital, RF and communication systems. Key services include signal integrity testing, rise/fall time measurements, eye diagram analysis, TDR (Time Domain Reflectometry) and TDT(Time Domain Transmissometry).We support applications in PCB debugging and high-frequency circuit characterization.

Measurement Capabilities:

- High-Speed Signal Analysis
- Wide Bandwidth Support, 6 GHz maximum bandwidth
- TDR (Time-Domain Reflectometry) and TDT (Time Domain Transmissometry)
- Pulse & Edge Measurements
- Eye Diagram Analysis

Antenna Test System: Our capabilities cover far field testing (0.6-50GHz), near field testing (0.6-50GHz), millimeter wave near field testing (supporting high-frequency antennas up to 110GHz) and on-chip antenna characterization (evaluating integrated antennas in ICs with high precision).

The test functions we can provide include

- Cylindrical/Planar Near-Far Field Transformation Function: Convert near-field amplitude-phase data to far-field radiation patterns.
- Antenna Directivity: Coefficient calculation, antenna gain analysis.
- Data Export Function: Export near-field measurement data and transformed far-field data.
- Radiation Pattern Display Function: Supports rectangular and polar coordinate displays for 2D patterns, as well as 3D pattern intensity maps, contour plots, 3D surface plots, and polar sphere plots.
- 2D Pattern Analysis Function: Automatically provides main lobe position and level, side lobe position and level, n-dB beamwidth, null position and level, etc.
- Normalization Function: Normalize radiation patterns, supporting fixed-value normalization and self-normalization.
- Multi-Curve Display Function: Simultaneous display and comparison of multiple radiation patterns and cross-sections.
- Supports automatic loading of measurement data, pre-transformation data cropping, and transformation to multiple polarizations and far-field coordinate systems.

(1) Antenna Near/Far Field (0.6-50GHz) Testing System Configuration

- Chamber Dimensions: 6m (L) × 3m (W) × 3m (H)
- Test Equipment: Vector Network Analyzer (VNA) : 600 MHz – 50 GHz
- Probe & Standard Horn Antenna
- Mechanical Setup:
 - 4-Axis Positioner (Azimuth, Elevation, Polarization; Repeatability $\leq 0.1^\circ$)
 - 4-Axis Scanner (X, Z, Y-axis repeatability: 0.05 mm; P-axis repeatability: 0.1°)
 - Scanner Controller
- Measurement Software: Real-time measurement software & test analysis software
- Test Accessories: Low-loss cables, coaxial adapters

Antenna Near/Far Field (0.6-50GHz) Testing System Specifications

Frequency Range (GHz)	0.6 – 50
Gain Measurement Uncertainty (dB)	± 0.5 (Near Field); ± 0.5 (Far Field)
-15 dB Sidelobe Level Uncertainty (dB)	± 1 (Near Field); ± 1.5 (Far Field)
-25 dB Sidelobe Level Uncertainty (dB)	± 2 (Near Field); ± 2.5 (Far Field)
-35 dB Sidelobe Level Uncertainty (dB)	± 3 (Near Field); ± 3.5 (Far Field)
Beam Pointing Accuracy (degree)	$\leq \text{Beamwidth} \times 1\%$ (Beamwidth $\geq 2^\circ$)
Radiation Pattern Display Range (degree)	± 180
Maximum Antenna Aperture (m)	0.5
Pattern Coordinate Types	Rectangular, Polar

(2) Antenna Near Field (75-110GHz) Testing System Configuration

- Chamber : 75 GHz – 110 GHz
- Test Equipment:
 - Vector Network Analyzer (VNA) 10 MHz – 26.5 GHz
 - Millimeter-Wave Frequency Extension 75 GHz – 110 GHz
 - Receive Probe 75 GHz – 110 GHz

Antenna Near/Far Field (0.6-50GHz) Testing System Specifications

Frequency Range (GHz)	75 – 110
Supported Polarizations	Linear, Circular
Radiation Pattern Display Range (degree)	± 180
Maximum Antenna Aperture (m)	0.5
Pattern Coordinate Types	Rectangular, Polar
Planar Scanning	Width Range: 0.9 m Height Range: 0.9 m Depth Range (Front-Back): 0.58 m (manually extendable to 0.85 m)
Cylindrical Scanning	Maximum Scan Radius: 0.86 m Height Range: 0.9 m AUT Horizontal Rotation: 360°
Spherical Scanning	Maximum Scan Radius: 0.86 m AUT Horizontal Rotation: 360° AUT Vertical Rotation: 360°

(3) Antenna on Chip (18-500GHz) Test System Configuration :

- Chamber Dimensions: 3m (L) × 3m (W) × 3m (H), featuring cleanroom-grade absorbing materials
- Test Equipment:
 - Vector Network Analyzer (VNA) with S-parameter modules (18GHz-500GHz)
 - Power amplifier
 - Duplexer module
 - Probe and standard horn antenna
- Feeding Method: Waveguide probe
- Probing Platform:
 - Precision probe station with 0.05-1mm positioning accuracy
 - 3-axis adjustable sample stage
 - 4-dimension adjustable probe holder
 - Vacuum-chuck sample holder
 - Pneumatic anti-vibration system for sample stage
 - 800× movable digital microscope
- Mechanical System:
 - Automated controller with dual-axis tilt positioner for E-plane/H-plane measurements
- Measurement Software:
 - Real-time measurement and test analysis software
- Test Accessories:
 - Low-loss cables
 - Waveguide and coaxial adapters

Antenna on Chip (18-500GHz) Test System Specifications

Operating Frequency Range (GHz)	18 - 500
Gain Measurement Uncertainty (dB)	±0.5
-15 dB Sidelobe Level Uncertainty (dB)	±1.5
-25 dB Sidelobe Level Uncertainty (dB)	±2.5
-35 dB Sidelobe Level Uncertainty (dB)	±3.5
Beam Pointing Accuracy (degree)	≤ Beamwidth × 1% (Beamwidth ≥ 2°)
Radiation Pattern Display Range (degree)	±180
Maximum Antenna Aperture (m)	0.5
Pattern Coordinate Types	Rectangular, Polar

mmWave/THz Circuit Test Platform: The mmWave/THz circuit test platform is designed for high-frequency characterization of integrated circuits and devices. Offering on-wafer measurements up to sub-THz frequencies, it features advanced RF probing, high resolution microscope and EMI shielding. The platform measures key parameters such as S-parameter, noise figure and power measurements, ensuring the performance of 5G, radar and terahertz system.

Measurement Capabilities

- S-parameter testing (up to 110 GHz or higher) for accurate characterization of scattering parameters (S11, S12, S21, S22) to evaluate impedance matching, insertion loss and isolation.
- Power spectrum analysis to measure signal power and harmonics across wide frequency bands.
- High dynamic range and low noise floor for precise signal integrity measurement.
- Support for both frequency-domain (VNA-based) and time-domain (oscilloscope) measurements.

Contact Us

For inquiries or to discuss your specific requirements, please reach out to us:

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